

Date: — 25/08/2025 — TASK 3.1

Using CLauses, Operators
And Function in queries

AIM:— To implement + of DML commands using
Clauses, operators and functions in queries

CLAUSES

→ Where, ORDER BY, GROUP BY, HAVING, DISTINCT

OPERATORS

→ equal(=)

→ BETWEEN

→ AND

→ AND

→ OR

→ IN

CREATE TABLE DEPARTMENTC

DEPTID INT PRIMARY KEY,
DEPTNAME VARCHAR(50) UNIQUE NOT NULL,
LOCATION VARCHAR(50) NOT NULL;

CREATE TABLE STUDENT 1C

STUDENT INT PRIMARY KEY,
NAME VARCHAR(50) NOT NULL.
AGE INT CHECK (AGE >= 18);
DEPTID INT FOREIGN KEY REFERENCES
DEPARTMENT(DEPTID),
CITY VARCHAR(50) DEFAULT 'UNKNOWN',
JOINDATE DATETIME DEFAULT GET DATE();

INSERT INTO DEPARTMENT VALUES

(1, 'CSE', 'HYDERABAD')

(2, 'ECE', 'MUMBAI');

(3, 'MECH', 'DELHI');

INSERT INTO STUDENT VALUES

(101, UPPER('RAHUL'), 20, 1, 'HYDERABAD');

INSERT INTO STUDENT VALUES

(102, 'ANJALI', 22, 2, 'MUMBAI');

INSERT INTO STUDENT VALUES

(103, 'KIRAN', 19, 1, 'PUNE');

INSERT INTO STUDENT VALUES

(104, 'MOHITH', 23, 3, 'DELHI');

INSERT INTO STUDENT VALUES

(105, 'SARAKHAN', 21, 1, 'HYDERABAD');

SELECT * FROM STUDENT;

	STUDENT ID	NAME	AGE	DEPTID	CITY	JOIN DATE
1	101	RAHUL	20	1	HYDERABAD	2025-08-26
2	102	ANJALI	22	2	MUMBAI	2025-08-26
3	103	KIRAN	19	1	PUNE	2025-08-26
4	104	MOHITH	23	3	DELHI	2025-08-26
5	105	SARAKHAN	21	1	HYDERABAD	2025-08-26

SELECT * FROM DEPARTMENT

	DEPTID	DEPTNAME	LOCATION
1	1	CSE	HYD
2	2	ECE	MUMBAI
3	3	MECH	DELHI

SELECT NAME, AGE

FROM STUDENT

WHERE AGE BETWEEN 19 and 22

	NAME	AGE
1	RAHUL	20
2	ANJALI	22
3	KIRAN	19
4	SARAKHAN	21


```
SELECT NAME, DEPTID
FROM STUDENT
WHERE DEPTID IN (1,3)
ORDER BY DEPTID DESC;
```

	NAME	DEPTID
1	MOHITH	3
2	SARAKHAN	1
3	RAHUL	1
4	KIRAN	1

```
UPDATE STUDENT
SET AGE = AGE + 1
WHERE DEPTID = 1 AND AGE < 21;
```

	STUDENT ID	NAME	AGE	DEPT-ID	CITY	JOINDATE
1	101	RAHUL	21	1	HYDERABAD	2025-8-26
2	102	ANJALI	22	2	MUMBAI	2025-8-26
3	103	KIRAN	20	1	PUNE	2025-8-26
4	104	MOHITH	23	3	DELHI	2025-8-26
5	105	SARAKHAN	21	1	HYDERABAD	2025-8-26

```
SELECT DISTINCT CITY
FROM STUDENT;
```

	CITY
1	DELHI
2	HYDERABAD
3	MUMBAT
4	PUNE

```
SELECT DEPTID, COUNT(*) AS TOTAL-STUDENTS
FROM STUDENT
GROUP BY DEPTID;
```

	DEPT-ID	TOTALSTUDENTS
1	1	3
2	2	1
3	3	1

```
SELECT DEPTID, COUNT(*) AS TOTAL-STUDENTS
FROM STUDENT
GROUP BY DEPTID
HAVING COUNT(*) >= 2;
```

	DEPTID	Total-stu
1	1	3

VELTECH	
EX No.	31
PERFORMANCE (5)	8
RESULT AND ANALYSIS (5)	8
VIVA VOCE (5)	8
RECORD (5)	8
TOTAL (20)	32
DATE	28/10/20

RESULT: The implementation of the clauses, operators & functions in the query (DDL and DML commands)

date: 25/08/25 TASK (3.2) AGGREGATE FUNCTIONS

AIM: To study & implement aggregate functions (COUNT(), SUM(), AVG(), MIN(), MAX()) on a sample database

AGGREGATE FUNCTIONS

They're mostly used with GROUPED BY to the rows

→ COUNT()

→ SUM()

→ AVG()

→ MIN()

→ MAX()

CREATE TABLE STUDENT2 (

ROLLNO INT PRIMARY KEY,

NAME VARCHAR(50),

AGE INT,

DEPTID INT,

MARKS INT);

INSERT INTO STUDENT2 VALUES

(1) ('Arun', 20, 101, 85),

(2) ('Shoha', 21, 101, 40),

(3) ('Ravi', 19, 102, 95),

(4) ('Priya', 22, 102, 95),

(5) ('Kiran', 20, 101, 60),

(6) ('Anita', 23, 103, 88);

SELECT * FROM STUDENT 2;

	ROLLNO	NAME	AGE	DEPT ID	MARKS
1	1	Ashish	20	101	85
2	2	Sneha	21	101	90
3	3	Ravi	19	102	70
4	4	Priya	22	102	95
5	5	Kiran	20	101	60
6	6	Anita	23	103	88

SELECT DEPT ID, AVG (MARKS) AS AVG - MARKS
FROM STUDENT 2
GROUPED BY DEPT ID;

	DEPT ID	AVG - MARKS
1	101	78
2	102	82
3	103	88

SELECT DEPT ID, MAX (MARKS) AS TOP - MARK
FROM STUDENT 2
GROUP BY DEPT ID;

	DEPT ID	TOP - MARK
1	101	90
2	102	95
3	103	88

SELECT DEPT ID, MIN (MARKS) AS LEAST MARK
FROM STUDENT 2
GROUP BY DEPT ID

	DEPT ID	LEAST MARKS
1	101	60
2	102	70
3	103	88

SELECT DEPTID, COUNT(*) AS STU_COUNT
FROM STUDENT2
GROUP BY DEPTID,

	DEPT ID	STU-COUNT
1	101	3
2	102	2
3	103	1

VELTECH	
EX No.	3.07
PERFORMANCE (5)	8
RESULT AND ANALYSIS (5)	8
VIVA VOCE (5)	3
RECORD (5)	1
TOTAL (20)	13
SIGN WITH DATE	28/8/22

RESULT: Implementation of all aggregate functions has been performed successfully on a table.